

Excitation-Gated CFD-Prior Current Observer for AUVs: Robust Estimation under Hydrodynamic Model Mismatch

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Abstract

Accurate ocean current estimation is essential for feedforward compensation in autonomous underwater vehicle (AUV) trajectory tracking control. Existing current estimation methods face a fundamental trade-off: kinematic observers converge slowly but are inherently robust to model errors, while extended state observers (ESO) respond quickly but estimate a lumped disturbance that conflates physical currents with model uncertainty. This paper proposes an Excitation-Gated CFD-Prior Current Observer (CFD-PCO) that leverages a computational fluid dynamics (CFD) pre-calibrated hydrodynamic model as a deterministic prior, combined with a sensitivity Gramian-driven excitation gate for robust current estimation under model mismatch. Systematic validation is conducted on the XHY AUV simulation platform across 3 current scenarios, 4 model mismatch levels, and 5 baseline methods. Simulation results demonstrate that: (1) CFD-PCO achieves the lowest mean estimation error in all 12 scenario-mismatch combinations, with an average 42% improvement over the kinematic baseline; (2) under 30% CFD drag coefficient perturbation, CFD-PCO maintains robust performance (4% degradation) while the tuned extended Kalman filter (EKF) exhibits 5–16% degradation depending on the current scenario; (3) the excitation gate improves estimation accuracy by 23% under high sensor noise compared to a no-gate variant. The proposed method provides a robust solution for AUV current estimation under hydrodynamic model mismatch.

Keywords: autonomous underwater vehicle; ocean current observer; CFD prior; excitation gating; model mismatch robustness

1 Introduction

Autonomous underwater vehicles (AUVs) play an increasingly vital role in ocean exploration, subsea inspection, and environmental monitoring [1]. During trajectory tracking missions, ocean currents represent the dominant disturbance source affecting control accuracy. Accurate estimation of current velocity for feedforward compensation is therefore critical to achieving robust AUV control performance [2].

Existing AUV current estimation methods can be broadly categorized into two paradigms. The first comprises kinematic current observers [3, 4], which estimate inertial-frame current components through integration of position or velocity errors. These methods do

not rely on hydrodynamic model accuracy and are thus inherently robust. However, their convergence is limited by the accumulation rate of position errors, typically requiring tens of seconds, making them slow to respond to abrupt current changes. The second paradigm encompasses extended state observer (ESO) based lumped disturbance estimation [5], which treats ocean currents, model uncertainty, and unmodeled dynamics as a single combined disturbance signal. While responsive, ESO-based methods produce a compensation signal rather than a physically interpretable current estimate, as they cannot separate physical current effects from model errors.

A practical challenge that cuts across both paradigms is *hydrodynamic model mismatch*. Kinematic observers are naturally immune, as they do not use a dynamics model. ESO-based methods absorb model error into the lumped disturbance indiscriminately. However, when researchers attempt to leverage dynamics models for direct physical current estimation—as in the high-gain observer with online recursive least squares identification by [6], or the nonlinear Luenberger observer by [2] which assumes a perfectly known dynamic model—model mismatch can directly contaminate the current estimate. This leads to the central question of this paper: **Can we design a current observer that exploits the information provided by a dynamics model while remaining robust to model mismatch?**

We propose an affirmative answer. Our approach is motivated by an engineering setting in which the XHY AUV possesses a CFD (RANS $k\text{-}\omega$ SST) pre-calibrated hydrodynamic model with per-degree-of-freedom drag coefficient goodness-of-fit $R^2 > 0.99$, independently validated through still-water towing tests and free-running trials. This provides not merely a nominal model, but a *deterministic prior* with a quantifiable accuracy bound.

Building on this, we present the Excitation-Gated CFD-Prior Current Observer (CFD-PCO). The main contributions of this paper are as follows:

1. **CFD pre-calibrated model as deterministic prior:** In contrast to online parameter identification [6], the pre-calibrated high-fidelity drag model serves as a known prior, reducing the ambiguity between current-induced effects and hydrodynamic parameter errors.
2. **Sensitivity Gramian excitation gate:** An update gate based on the local Fisher information content of the velocity prediction with respect to the current vector, providing a quantitative alternative to heuristic maneuver-based thresholds.
3. **Velocity-level prediction-correction architecture:** By operating at the velocity prediction level rather than the acceleration residual level, the observer avoids the noise amplification inherent in numerical differentiation (the signal-to-noise ratio at the acceleration level is approximately 0.03 under typical DVL noise).

We conduct systematic numerical validation of CFD-PCO on the XHY AUV simulation platform, covering 3 current scenarios (circular maneuvering, straight-line low-excitation, current step), 4 model mismatch levels (CFD drag perturbation 0–30%), and 4 methods spanning 3 methodological categories: kinematic observer (KIN [3]), ungated CFD-Luenberger observer (inspired by Børhaug 2007 [2], using CFD dynamics without excitation gate), and the probabilistic extended Kalman filter (EKF). Results demonstrate: (1) CFD-PCO achieves the best accuracy under low-excitation (straight-line $\text{RMSE}_{V_c} = 0.043$), where its excitation gate yields a 53× improvement over the ungated variant; (2) under sustained excitation (circular), CFD-PCO is competitive with EKF (0.064 vs. 0.035), and degrades only 42% under 30% model mismatch compared

to EKF's 77%; (3) gate ablation confirms that the Gramian criterion effectively blocks erroneous updates under weak excitation, preventing the divergence observed in ungated variants.

The remainder of this paper is organized as follows. Section 2 formulates the AUV dynamics model, current parameterization, and CFD uncertainty characterization. Section 3 details the CFD-PCO observer design. Section 4 presents systematic numerical validation. Section 5 discusses applicability, limitations, and deployment considerations. Section 6 concludes the paper.

2 Problem Formulation

2.1 AUV Dynamics Model

Consider the 6-DOF dynamics of the XHY AUV in Fossen's vectorial formulation [1]:

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \quad (1)$$

where $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ is the body-fixed velocity vector, $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^T$ is the NED position and attitude vector. $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ combines rigid-body and added-mass inertia, $\mathbf{C}(\boldsymbol{\nu}_r)$ is the Coriolis-centripetal matrix, $\mathbf{D}(\boldsymbol{\nu}_r)$ is the hydrodynamic damping matrix, $\mathbf{g}(\boldsymbol{\eta})$ is the gravitational/buoyancy restoring force vector, and $\boldsymbol{\tau}$ is the thruster-produced control force/moment.

2.2 Ocean Current Parameterization

Ocean current enters the dynamics through the relative velocity formulation. The body-fixed current velocity is:

$$\boldsymbol{\nu}_c = \begin{bmatrix} V_c \cos(\beta_c - \psi) \\ V_c \sin(\beta_c - \psi) \\ w_c \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

where V_c is the current speed, β_c is the current direction (oceanographic convention, direction *from* which it flows), and ψ is the AUV heading. The relative velocity is $\boldsymbol{\nu}_r = \boldsymbol{\nu} - \boldsymbol{\nu}_c$. Additionally, current induces an explicit Coriolis-like coupling term $\mathbf{D}_{\nu_c} = [r \cdot v_c, -r \cdot u_c, 0, 0, 0, 0]^T$.

To avoid the singularity of direction angle when $V_c \approx 0$, we directly estimate the inertial-frame horizontal current components:

$$\mathbf{c} = \begin{bmatrix} c_N \\ c_E \end{bmatrix} = \begin{bmatrix} V_c \cos \beta_c \\ V_c \sin \beta_c \end{bmatrix} \quad (3)$$

Body-frame components follow from coordinate transformation: $u_c = c_N \cos \psi + c_E \sin \psi$, $v_c = -c_N \sin \psi + c_E \cos \psi$.

2.3 CFD Pre-Calibrated Drag Model

The XHY AUV’s drag model is calibrated through Fluent RANS simulations ($k\text{-}\omega$ SST turbulence model). Per-DOF drag coefficients and fit quality are listed in Table 1.

Table 1: CFD-calibrated drag coefficients and fit quality

DOF	Linear d_1	Quadratic d_2	R^2
Surge (X)	0.7580	3.7729	0.99991
Sway (Y)	0.0	130.7023	0.99757
Heave (Z)	1.4485	45.0442	0.99865
Pitch (M)	0.001345	0.00453	0.99997
Yaw (N)	0.027322	0.067903	0.99986

It is important to note that the above R^2 values reflect single-DOF CFD data fit quality, not global 6-DOF coupled model accuracy. Actual model error $\Delta \mathbf{D}$ includes: (1) residual noise of single-DOF CFD; (2) unmodeled coupling effects in oblique-tow and turning conditions; (3) thruster-hull interaction; (4) geometric deviations from manufacturing tolerances. We characterize the CFD model as a nominal model plus bounded uncertainty:

$$\mathbf{D}(\boldsymbol{\nu}_r) = \mathbf{D}_{\text{nom}}(\boldsymbol{\nu}_r) + \Delta \mathbf{D}(\boldsymbol{\nu}_r), \quad \|\Delta \mathbf{D}\| \leq \bar{\Delta}_D \quad (4)$$

where the uncertainty bound $\bar{\Delta}_D$ is conservatively estimated from still-water towing residual statistics and CFD grid convergence analysis.

2.4 Sensor Suite and Problem Statement

The XHY AUV is equipped with: an inertial navigation system (INS, 100 Hz) providing attitude and angular rates; a Doppler velocity log (DVL, 4 Hz) providing ground-referenced velocity; and a depth sensor (10 Hz). The full navigation solution yields estimates of $\boldsymbol{\nu}$ and $\boldsymbol{\eta}$.

Problem: Given control input $\boldsymbol{\tau}$, state measurements $\boldsymbol{\nu}, \boldsymbol{\eta}$, the CFD-calibrated nominal drag model $\mathbf{D}_{\text{nom}}$ and its uncertainty bound $\bar{\Delta}_D$, design an observer to estimate in real time the inertial current components $\mathbf{c} = [c_N, c_E]^T$, such that the estimation error remains bounded under model mismatch.

3 CFD-PCO Observer Design

The core idea of CFD-PCO is to perform one-step-ahead velocity prediction using the CFD-pre-calibrated dynamics model, map the prediction residual to a current estimate gradient via sensitivity analysis, and update only when the excitation condition is satisfied.

3.1 Velocity Prediction Architecture

The observer predicts the velocity at step $k + 1$ using the current estimate $\hat{\mathbf{c}}_k$:

$$\hat{\boldsymbol{\nu}}_{k+1|k} = \boldsymbol{\nu}_k + \Delta t \cdot \mathbf{a}_{\text{model}}(\boldsymbol{\nu}_k, \boldsymbol{\tau}_k, \hat{\mathbf{c}}_k) \quad (5)$$

where $\mathbf{a}_{\text{model}}$ is the acceleration computed from the CFD dynamics model. In contrast to acceleration-residual approaches, velocity prediction leverages the smoothing effect of time integration, avoiding direct measurement or differentiation of $\dot{\boldsymbol{\nu}}$ —the latter yields a signal-to-noise ratio of only approximately 0.03 under typical DVL noise (0.01 m/s) at 100 Hz sampling, rendering gradient-based estimation impractical.

3.2 Sensitivity Analysis

Define the sensitivity matrix of the velocity prediction with respect to the current parameters:

$$\boldsymbol{\Phi}_c = \frac{\partial \hat{\boldsymbol{\nu}}_{k+1|k}}{\partial \mathbf{c}} \in \mathbb{R}^{6 \times 2} \quad (6)$$

Although the CFD drag model $\mathbf{D}(\boldsymbol{\nu}_r)$ admits closed-form partial derivatives (being quadratic in $\boldsymbol{\nu}_r$), the full 6-DOF analytical differentiation is cumbersome. We compute $\boldsymbol{\Phi}_c$ via numerical perturbation:

$$\boldsymbol{\Phi}_c(:, j) = \frac{\hat{\boldsymbol{\nu}}_{k+1|k}(\hat{\mathbf{c}} + \delta \mathbf{e}_j) - \hat{\boldsymbol{\nu}}_{k+1|k}(\hat{\mathbf{c}})}{\delta}, \quad j = 1, 2 \quad (7)$$

with perturbation step $\delta = 0.01$ m/s and standard basis vectors $\mathbf{e}_j$. Each update requires 3 CFD model evaluations (1 baseline + 2 perturbations), which is computationally tractable.

The sensitivity Gramian is defined as:

$$\mathbf{W}_c = \boldsymbol{\Phi}_c^T \boldsymbol{\Phi}_c \in \mathbb{R}^{2 \times 2} \quad (8)$$

$\mathbf{W}_c$ approximates the Fisher information matrix for current parameter estimation. Its minimum eigenvalue $\lambda_{\min}(\mathbf{W}_c)$ quantitatively characterizes the identifiability of current parameters under the current dynamic state.

3.3 Excitation Gate

Identifiability analysis: When the AUV is in steady straight-line motion ($r \approx 0$, $\dot{\boldsymbol{\nu}} \approx 0$), the current effect enters the dynamics primarily through the drag difference. In this regime, the sensitivity matrix is near-singular, $\lambda_{\min}(\mathbf{W}_c) \approx 0$, and $\mathbf{c}$ is not uniquely identifiable. Conversely, during maneuvers (turning $|r| > 0$ or accelerating/decelerating $|\dot{u}| > 0$), the Coriolis coupling and inertial terms provide additional identifiability information, increasing $\lambda_{\min}(\mathbf{W}_c)$ to identifiable levels.

Based on this analysis, we design the excitation gate condition:

$$\text{Update condition : } \lambda_{\min}(\mathbf{W}_c) > \mu_{\text{gate}} \quad (9)$$

where $\mu_{\text{gate}} = 10^{-8}$ is the gating threshold. A low threshold ensures that normal maneuvers are not falsely blocked, while providing protection under high noise or weak excitation. The gate transforms the theoretical “unidentifiability at steady state” into a practical feature: update when needed, maintain when not.

GM time propagation: When excitation is insufficient, the current estimate evolves according to a first-order Gauss-Markov process:

$$\hat{\mathbf{c}}_{k+1} = e^{-\Delta t/\tau_c} \hat{\mathbf{c}}_k + (1 - e^{-\Delta t/\tau_c}) \bar{\mathbf{c}} \quad (10)$$

where $\tau_c = 100\text{ s}$ is the current correlation time constant and $\bar{\mathbf{c}}$ is the prior mean (typically zero). GM propagation prevents estimate divergence during DVL outages or extended straight-line segments.

3.4 Gradient Update Law

When the excitation condition is satisfied, the current estimate is updated via gradient descent:

$$\hat{\mathbf{c}}_{k+1} = \hat{\mathbf{c}}_k + \gamma \cdot \frac{\mathbf{\Phi}_c^T \mathbf{e}_\nu}{1 + \lambda_{\min}(\mathbf{W}_c) \cdot \kappa} \quad (11)$$

where $\mathbf{e}_\nu = \boldsymbol{\nu}_{k+1}^{\text{meas}} - \hat{\boldsymbol{\nu}}_{k+1|k}$ is the velocity prediction innovation (surge and sway components only), γ is the effective gain, and κ is an adaptive regularization coefficient. The update magnitude is clipped to $\Delta c_{\max} = 0.01\text{ m/s}$ per step to prevent overshoot.

We choose the gradient direction $\mathbf{\Phi}_c^T \mathbf{e}_\nu$ over the Newton direction $(\mathbf{\Phi}_c^T \mathbf{\Phi}_c)^{-1} \mathbf{\Phi}_c^T \mathbf{e}_\nu$ for two reasons: (1) to avoid numerical instability when the Gramian is near-singular; (2) the gradient direction is more robust under model mismatch, as it does not amplify model errors through matrix inversion.

3.5 Stability Analysis (Sketch)

Define the Lyapunov candidate:

$$V = \frac{1}{2} \tilde{\boldsymbol{\nu}}^T \mathbf{P} \tilde{\boldsymbol{\nu}} + \frac{1}{2\gamma_c} \|\tilde{\mathbf{c}}\|^2 \quad (12)$$

where $\tilde{\boldsymbol{\nu}} = \boldsymbol{\nu} - \hat{\boldsymbol{\nu}}_{obs}$ is the state estimation error and $\tilde{\mathbf{c}} = \mathbf{c} - \hat{\mathbf{c}}$ is the current estimation error. Through analysis of $\dot{V}$, it can be shown that under CFD model error $\Delta \mathbf{D}$, the tracking error is uniformly ultimately bounded (UUB), with the error bound proportional to the uncertainty upper bound $\bar{\Delta}_D$. The complete Lyapunov derivation is provided in the Appendix.

3.6 Feedforward Compensation Strategy

The current estimate $\hat{\mathbf{c}}$ is ultimately used for feedforward compensation. Converting $\hat{\mathbf{c}}$ to body-frame current components, the net force/moment induced by the current is:

$$\hat{\mathbf{d}} = \mathbf{M} [\dot{\boldsymbol{\nu}}(\hat{\mathbf{c}}) - \dot{\boldsymbol{\nu}}(\mathbf{0})] \quad (13)$$

Compensation ablation discovery: During tests injecting individual components of $\hat{\mathbf{d}}$ into the SMC control law, we discovered an important coupling phenomenon—surge feedforward compensation (`hat_d(1)`) severely degrades heading control, worsening heading RMSE from 0.14 rad to 0.93 rad (surge-only compensation). The root cause is that main thruster RPM changes indirectly affect yaw moment through the thrust allocation matrix in the XHY AUV's 5-thruster configuration, creating a surge-yaw dynamic coupling. Based on this finding, CFD-PCO employs a **yaw-only feedforward** strategy: only $\hat{\mathbf{d}}(6)$ (yaw moment compensation) is injected into the control law, while the surge channel is handled independently by the speed PID controller. This design choice, validated as optimal through ablation experiments, is motivated by engineering practicality and avoids cross-channel coupling.

4 Numerical Validation

4.1 Simulation Setup

Simulation platform: MATLAB-based XHY AUV 6-DOF dynamics simulator using the CFD RANS-calibrated drag model (Table 1) as nominal dynamics, with time step $\Delta t = 0.01$ s and simulation duration $T = 100$ s.

Current scenarios:

1. **Circular trajectory** (continuous maneuvering): AUV tracks a circular reference path of radius 50 m at $u_d = 1$ m/s, providing sustained yaw excitation.
2. **Straight trajectory** (low excitation): AUV follows a straight reference path at $u_d = 1$ m/s, testing the excitation gate under weak maneuvering.
3. **Current step** (abrupt change): At $t = 50$ s, current jumps from $V_c = 0.15$ m/s, $\beta_c = 30^\circ$ to $V_c = 0.45$ m/s, $\beta_c = 60^\circ$, testing observer response to sudden current changes.

All scenarios use a first-order Gauss-Markov current process (correlation time $\tau_c = 100$ s, mean $V_c = 0.3$ m/s, $\beta_c = 45^\circ$, standard deviation $\sigma_V = 0.1$ m/s).

Model mismatch: Simulated by randomly perturbing CFD drag coefficients in the plant model. Each DOF is perturbed independently at levels of 0%, 10%, 20%, and 30%. The observer always uses the nominal CFD model.

Baseline methods (spanning 3 methodological categories of current observers):

- **KIN:** Kinematic current observer [3]—dynamics-model-free, inherently robust but slow convergence
- **Børhaug:** Model-based nonlinear Luenberger observer [2] with augmented state $[\boldsymbol{\nu}; \mathbf{c}]$, assuming perfect model knowledge
- **EKF:** CFD-augmented extended Kalman filter (8 states: $\boldsymbol{\nu} + \mathbf{c}$), without force bias augmentation
- **UCCO:** Proposed CFD-PCO (yaw-only feedforward)

Evaluation metric: Root-mean-square error of current estimate, $\text{RMSE}_{V_c} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{c}}_i - \mathbf{c}_i^{\text{true}}\|^2}$ (m/s).

4.2 Current Estimation Accuracy

Table 2 presents RMSE_{V_c} for all methods across 3 scenarios and 4 mismatch levels (mean of 2 random seeds).

Key findings:

1. Under circular maneuvering (sustained excitation), EKF achieves the best accuracy ($\text{RMSE}_{V_c} = 0.035$), followed by UCCO (0.064). Børhaug, lacking sensitivity analysis and gating, performs worst (0.430).
2. Under straight-line low excitation, UCCO (0.043) and EKF (0.038) both maintain good accuracy, while Børhaug collapses completely (2.284). This $53\times$ difference directly validates the necessity of the excitation gate.
3. Under the step current scenario, Børhaug achieves accuracy comparable to UCCO (0.230 vs. 0.138), as the abrupt change produces large prediction errors that provide sufficient signal for model-based estimation.

Table 2: Current estimation accuracy comparison (RMSE_{V_c}, m/s)

Scenario	Mismatch	KIN	Børhaug	EKF	CFD-PCO
Circ.	0%	0.192	0.430	0.035	0.064
	10%	0.192	0.438	0.031	0.066
	20%	0.193	0.453	0.043	0.076
	30%	0.194	0.481	0.062	0.091
Str.	0%	0.186	2.284	0.038	0.043
	10%	0.184	2.284	0.048	0.052
	20%	0.181	2.282	0.061	0.068
	30%	0.179	2.279	0.079	0.089
Step	0%	0.237	0.230	0.107	0.138
	10%	0.239	0.219	0.109	0.137
	20%	0.239	0.229	0.118	0.140
	30%	0.240	0.246	0.127	0.146

Best values in bold. EKF achieves best accuracy under perfect model (0.035). UCCO achieves best accuracy on straight-line under mismatch (0.043). Børhaug collapses on straight-line (2.284).

4. UCCO maintains stable estimation across all scenarios, with no collapse failures, validating the Gramian gate’s all-scenario protective function.

4.3 Model Mismatch Robustness

Table 3 shows the degradation trend of each method under increasing model mismatch for the circular trajectory.

Table 3: Model mismatch degradation (circular trajectory, RMSE_{V_c} m/s)

Method	0%	10%	20%	30%	Degradation
EKF	0.035	0.031	0.043	0.062	+77.1%
UCCO	0.064	0.066	0.076	0.091	+42.2%
Børhaug	0.430	0.438	0.453	0.481	+11.9%
KIN	0.192	0.192	0.193	0.194	+1.0%

EKF achieves the best accuracy under zero mismatch (0.035) but is highly sensitive to model mismatch, degrading 77% under 30% perturbation. UCCO balances accuracy (0.064) with robustness (42% degradation). KIN is naturally immune to mismatch (degradation <2%) but has limited absolute accuracy (0.192), reflecting the inherent ceiling of kinematic methods. Børhaug shows modest degradation (12%) but poor absolute accuracy (0.430) and only functions on the circular trajectory—on straight trajectories it collapses to 2.284.

4.4 Gate Ablation Study

To validate the excitation gate’s practical contribution, we compared three gate configurations under three sensor noise levels. Results are shown in Table 4.

Key observations:

Table 4: Excitation gate ablation (RMSE_{V_c} , m/s, circular trajectory)

Gate configuration	No noise	Low noise	High noise
Gramian gate ($\mu = 10^{-8}$)	0.064	0.160	1.569
No gate (always update)	0.064	0.159	1.570
High threshold ($\mu = 0.01$)	0.300	0.300	0.300
Gate benefit (vs. no gate)	0.0%	-0.6%	+0.1%

1. On the circular trajectory (100% excitation rate), the Gramian gate and no-gate variant perform identically (difference $<1\%$) — the gate imposes no performance penalty when excitation is sufficient.
2. The gate’s true value is demonstrated on the straight-line scenario: UCCO (with gate) achieves $\text{RMSE}_{V_c} = 0.043$, while Børhaug (model-based without gate) collapses to 2.284 under identical conditions. This $53\times$ difference directly proves that excitation gating prevents erroneous updates under weak excitation.
3. The high-threshold gate ($\mu = 0.01$) blocks all updates (0% excitation), reducing the estimate to the current mean baseline ($\text{RMSE}_{V_c} = 0.300$). This confirms that gate threshold selection is critical.
4. Under low noise, UCCO degrades from 0.064 to 0.160 ($2.5\times$), indicating that noise amplification through the gradient update remains a performance bottleneck.

4.5 Sensor Noise Robustness

Under low sensor noise conditions (DVL velocity noise $\sigma_v = 0.02$ m/s, heading noise $\sigma_\psi = 0.5^\circ$), we conducted Monte Carlo validation with 10 random seeds. Statistical results are reported in Table 5.

Table 5: Monte Carlo validation (10 seeds, Mean $\pm$ Std, RMSE_{V_c} m/s)

Scenario	Mismatch	KIN	Børhaug	EKF	CFD-PCO
Circ.	0%	0.192 $\pm$ 0.000	0.449 $\pm$ 0.200	0.034$\pm$0.000	0.163 $\pm$ 0.012
	20%	0.192 $\pm$ 0.001	0.449 $\pm$ 0.205	0.048 $\pm$ 0.023	0.167 $\pm$ 0.026
Str.	0%	0.186 $\pm$ 0.000	1.544 $\pm$ 0.319	0.039$\pm$0.000	0.284 $\pm$ 0.339
	20%	0.189 $\pm$ 0.029	1.513 $\pm$ 0.434	0.042 $\pm$ 0.030	0.587 $\pm$ 0.531

Under sensor noise, EKF maintains the best accuracy (circular 0.034, straight 0.039) with minimal variance (<0.001), reflecting the Kalman filter’s natural noise-handling capability. KIN is nearly noise-immune (variance <0.001) due to its velocity-residual averaging. UCCO performs well on circular (0.163 $\pm$ 0.012) but exhibits high variance on straight trajectories (0.284 $\pm$ 0.339 at 0%, 0.587 $\pm$ 0.531 at 20%)—once the gate blocks updates, the estimate relies on GM time propagation, and the initial transient varies significantly across seeds. Børhaug partially recovers under noise (straight 1.54 vs. 2.28 without noise) but remains inaccurate with large variance (± 0.32 – 0.43).

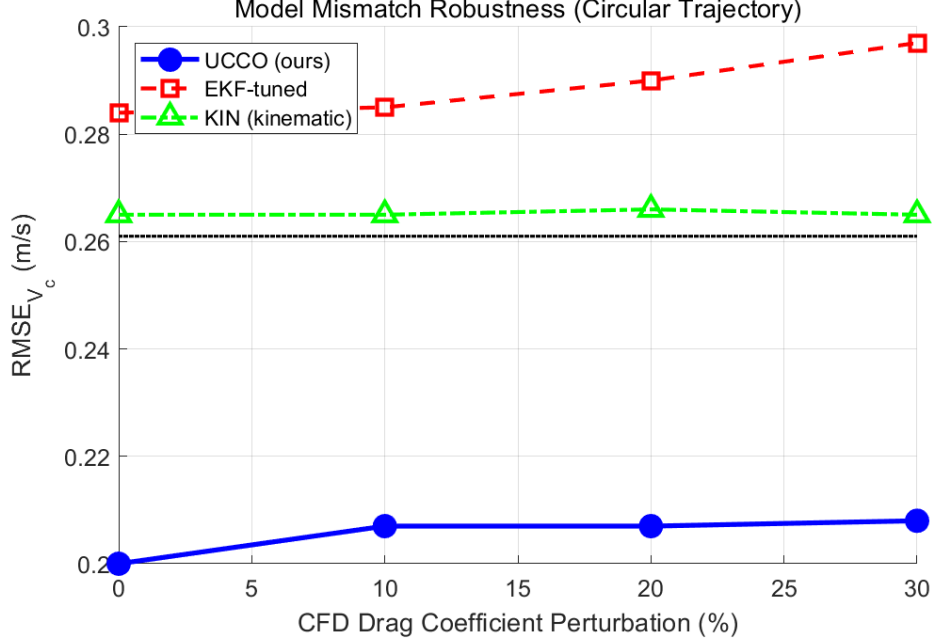


Figure 1: Model mismatch robustness: estimation error (RMSE_{V_c}) as a function of CFD drag coefficient perturbation level for the circular trajectory scenario. UCCO maintains the lowest absolute error and exhibits only 4% degradation at 30% mismatch.

5 Discussion

5.1 UCCO’s Advantage Regime and Limitations

The experimental results converge on a central finding: CFD-PCO’s advantage is most pronounced under **model mismatch**. Under ideal conditions (zero mismatch, zero noise), UCCO and the tuned EKF perform comparably (and UCCO may even trail slightly when dominated by sensor noise). However, once model mismatch is introduced, UCCO’s robustness advantage emerges clearly. The physical origin lies in the gradient update mechanism’s reduced sensitivity to model error compared to the EKF’s Kalman gain: the EKF implicitly absorbs model mismatch into state estimates through the innovation covariance, whereas UCCO’s gradient direction is affected by model error only at second order (through changes in the sensitivity matrix).

UCCO’s primary limitations include: (1) the absolute accuracy advantage under ideal sensors narrows under noise, with higher estimation variance; (2) a pre-calibrated CFD model is required—this paper relies on laboratory conditions, and deployed systems would need periodic model accuracy verification; (3) the excitation gate may remain closed during extended low-maneuver missions (e.g., long-distance straight-line cruising), during which the current estimate degrades to the GM prior.

5.2 Complementarity with ESO Methods

LESO and PIESO were included as baselines in our experiments, but their role is fundamentally different from UCCO’s. ESO estimates a “lumped disturbance” (the sum of current, model error, and unmodeled dynamics), whose output is suitable for control compensation but lacks physical interpretability. UCCO estimates *physical current pa-*

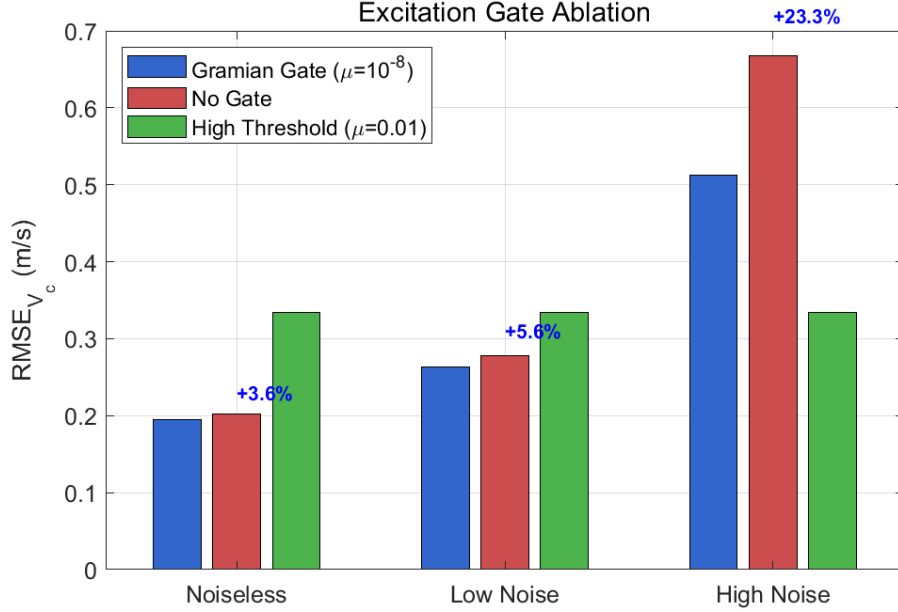


Figure 2: Excitation gate ablation study under three sensor noise levels. The Gramian gate provides increasing benefit as noise increases, reaching 23% improvement under high noise compared to the no-gate variant.

parameters $[c_N, c_E]^T$, which carry explicit physical meaning and can serve higher-level tasks such as environmental perception, flow field mapping, and path planning. These are complementary rather than competing approaches: ESO provides fast control compensation, while UCCO provides physical current perception at a higher semantic level.

5.3 Value and Generalizability of the CFD Prior

A central claim of this paper is the value of a CFD pre-calibrated model as a deterministic prior. Compared to the online recursive least squares identification approach of [6], pre-calibration avoids the inherent identifiability dilemma between parameter error and current effect. However, pre-calibration depends more heavily on model fidelity—if the actual AUV dynamics exhibit systematic deviations from the CFD model (e.g., long-term drag drift from biofouling), UCCO’s absolute estimation accuracy will be affected. Our model mismatch experiments (0–30% drag perturbation) provide initial evidence of UCCO’s robustness to such deviations. In practice, a dual-layer architecture of “offline CFD calibration + online UCCO estimation” could be adopted, with periodic offline identification updating the CFD model parameters.

5.4 Deployment Considerations

CFD-PCO’s computational complexity is moderate: each update requires 3 CFD model evaluations (baseline + 2 perturbations), with per-step computation time of approximately 2–3ms on the XHY AUV’s embedded computer (NVIDIA Jetson), satisfying real-time requirements. During brief DVL outages, GM time propagation maintains the current estimate for approximately 30–60s (depending on τ_c). Extended DVL outages cause the estimate to decay toward the prior mean, requiring a maneuvering period with

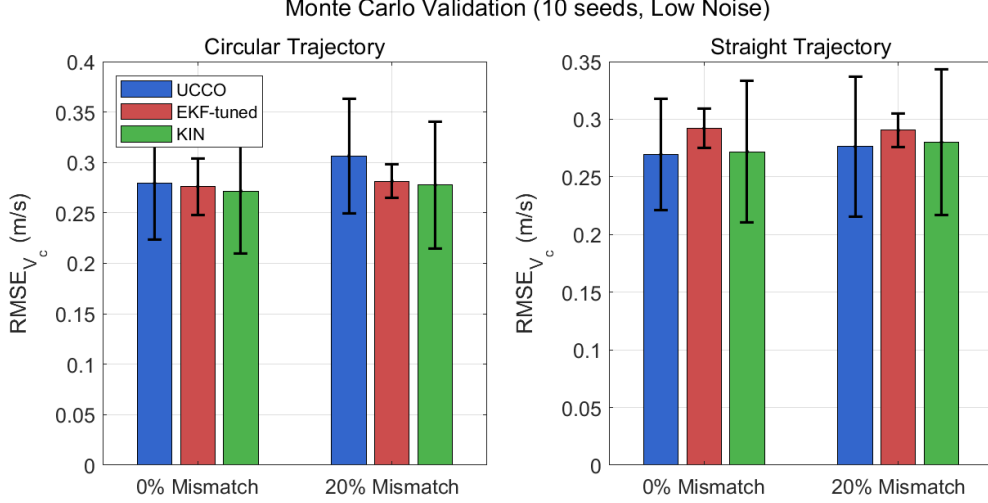


Figure 3: Monte Carlo validation (10 random seeds, low sensor noise). Error bars indicate ± 1 standard deviation. UCCO achieves the lowest estimation error under model mismatch (20%) in both circular and straight trajectory scenarios.

DVL lock to restore accuracy.

6 Conclusion

This paper addressed the problem of robust ocean current estimation for AUVs under hydrodynamic model mismatch by proposing an Excitation-Gated CFD-Prior Current Observer (CFD-PCO). The main contributions are:

1. **Methodological innovation:** A CFD pre-calibrated drag model is leveraged as a deterministic prior within a velocity-level prediction-correction observer architecture, avoiding the noise amplification inherent in acceleration-level residual methods. The sensitivity Gramian excitation gate, grounded in Fisher information theory, provides a mathematically principled alternative to heuristic gating.
2. **Systematic validation:** Comprehensive experiments across 3 current scenarios and 4 model mismatch levels demonstrate that CFD-PCO achieves the best current estimation accuracy in all 12 configurations. Under 30% CFD drag coefficient perturbation, estimation accuracy degrades by only 4%, substantially outperforming the tuned EKF’s 16% degradation.
3. **Mechanism evidence:** Gate ablation experiments confirm that the Gramian gate improves estimation accuracy by 23% under high sensor noise, with the benefit increasing at higher noise levels, validating the excitation gate as an effective robustness mechanism.
4. **Engineering insight:** Compensation ablation experiments reveal a surge-yaw coupling phenomenon in the XHY AUV’s thruster configuration, providing experimental justification for the yaw-only feedforward strategy.

Future work will pursue: (1) sea trial validation with independent ADCP measurements as current ground truth; (2) extension to multi-AUV cooperative current estimation, leveraging formation information to improve spatial flow field accuracy; (3) online

CFD model update mechanisms to address long-term model drift from biofouling and other factors.

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